Domains, DNS, Websites & SSL Certificates

Quick Reference Guide

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🌐 Domains

What is a Domain? A domain is a human-readable address that identifies a location on the internet, like `google.com` or `microsoft.com`.

Domain Structure

```
subdomain.domain.tld
www.example.com
|   |       |
|   |       └── Top Level Domain (.com, .org, .net)
|   └────────── Second Level Domain (your brand/company)
└────────────── Subdomain (www, mail, ftp)
```

Common TLDs (Top Level Domains)

- **.com** - Commercial (most popular)
- **.org** - Organizations
- **.net** - Network-related
- **.gov** - Government
- **.edu** - Educational institutions

🔍 DNS (Domain Name System)

What is DNS? DNS translates human-readable domain names into IP addresses that computers use to communicate.

How DNS Works (Simple Process)

1. **User types:** `www.example.com` in browser
2. **DNS resolver asks:** Root servers → TLD servers → Authoritative servers
3. **Response:** Returns IP address (e.g., 192.168.1.100)
4. **Browser connects:** To the actual web server using the IP address

Key DNS Record Types

Record Type	Purpose	Example
A	Points domain to IPv4 address	`example.com → 192.168.1.100`
AAAA	Points domain to IPv6 address	`example.com → 2001:db8::1`
CNAME	Points domain to another domain	`www.example.com → example.com`
MX	Mail server records	`mail.example.com`
NS	Name server records	Defines DNS servers for domain
TXT	Text records for verification	SPF, DKIM, domain verification

DNS Propagation

- **Time:** 24-48 hours globally
- **Reason:** DNS changes need to update across worldwide servers
- **TTL:** Time To Live setting controls cache duration

🖥️ Websites

What is a Website? A collection of web pages hosted on a web server, accessible via a domain name.

Website Components

- **Web Server:** Computer that hosts website files (Apache, Nginx, IIS)
- **Web Pages:** HTML, CSS, JavaScript files
- **Database:** Stores dynamic content (MySQL, PostgreSQL)
- **Content:** Text, images, videos, documents

How Websites Work

1. **User enters URL:** Browser requests website
2. **DNS lookup:** Finds server IP address
3. **HTTP/HTTPS request:** Browser contacts web server
4. **Server responds:** Sends HTML, CSS, images
5. **Browser renders:** Displays the webpage

Website Hosting Types

- **Shared Hosting:** Multiple sites on one server (cheapest)
- **VPS Hosting:** Virtual private server (more control)
- **Dedicated Hosting:** Entire server for one website (expensive)
- **Cloud Hosting:** Distributed across multiple servers (scalable)

🔒 SSL Certificates

What is SSL? SSL (Secure Sockets Layer) encrypts data between a user's browser and the web server, ensuring secure communication.

SSL Indicators

- **Green padlock** in browser address bar
- **https://** instead of http://
- **"Secure" or "Not Secure"** warning in modern browsers

Types of SSL Certificates

Type	Validation Level	Use Case	Cost
Domain Validated (DV)	Basic domain ownership	Personal websites, blogs	Low
Organization Validated (OV)	Domain + organization verification	Business websites	Medium
Extended Validation (EV)	Highest validation level	E-commerce, banking	High
Wildcard	Covers all subdomains	`*.example.com`	Medium-High

SSL Certificate Lifecycle

1. **Purchase/Obtain:** From Certificate Authority (CA)
2. **Install:** On web server
3. **Configure:** Web server to use HTTPS
4. **Monitor:** Expiration dates
5. **Renew:** Before expiration (typically 1-2 years)

Key Takeaway: Domains provide the address, DNS provides the directions, websites provide the content, and SSL certificates provide the security - all working together to deliver secure web experiences.